# Project Deliverable C:

# Design Criteria and Target Specifications

**Group 16: The Sample Snatchers**

Ang Li, Eden Irwin, Isaac Pinarski

Radhe Pandey, and Tiam Morrow-Rogers

Faculty of Engineering – University of Ottawa

GNG1103: Engineering Design

Dr. David Knox

February 2, 2025

# Table of Contents

1.0 Introduction....4
1.1 Introduction....4
1.2 Problem Statement....4
2.0 Design Criteria and Target Specifications....5
2.1 Defining Design Criteria....5
2.2 Technical and User Benchmarking....5
2.3 Engineering Design Specifications....7
3.0 Conclusion....8
3.1 General Progress....8
3.2 Client Meeting Reflection....8
3.3 Trello: Task Plan Update....8
Citations....9

# List of Tables

Table 2.1-1: Design Criteria....5
Table 2.2-1: Product Benchmarking....6
Table 2.2-2: User Benchmarking - Tasks of CWEST....6
Table 2.3-1: EDS - Functional Requirements....7
Table 2.3-2: EDS - Constraints....7
Table 2.3-3: EDS - Non-Functional Requirements....7

# Abstract

This report outlines, organizes, and analyzes this project's fundamental design criteria and target specifications. The project aims to develop a compact, fully removable device for extracting metal samples from nuclear reactor pressure tubes.

The document details the prioritized design criteria derived from user needs, incorporating functional and non-functional requirements and key constraints. Technical and user benchmarking have been conducted to investigate and deeply understand the user's potential missed needs and wants for the design. Engineering design specifications (EDS) for the categorized design criteria are established into quantifiable attributes derived from the prioritized needs list. These will be used later to benchmark the prototypes developed.

Conclusively, the report reflects on insights from the preliminary client meeting, and details updates contributing towards the project’s task management. The report's content will guide the development and evaluation of ideations, the following stage of the Design Thinking process.

# 1.0 Introduction

## 1.1 Introduction

This report centers around the define stage. It aims to create prioritized design criteria, execute technical and user benchmarking, and determine target specifications that will aid in developing our final solution. Previously developed needs will be used to create the necessary criteria for assessing developed solutions for the problem statement.

## 1.2 Problem Statement

Operators who service nuclear reactors need a device to collect metal samples from pressure tubes while minimizing exposure to contaminated material. The solution should be compact and completely removable from the pressure tube, and it should also be able to report the stage of operation within a limited timeframe.

# 2.0 Design Criteria and Target Specifications

## 2.1 Defining Design Criteria

Based on the team's list of interpreted user needs, we have defined the prioritized design criteria as exemplified in Table 2.1-1. This table will be referenced to build the Engineering Design Specifications (EDS) in Section 2.3. The criteria are organized based on the hierarchy of needs done in the previous week.

Table 2.1-1: Design Criteria

| Rank | Need | High Priority | Design Criteria |
|---|---|---|---|
| 1 | Sample is contained to limit exposure | SAFETY | Thickness (mm) Material (Density) |
| 2 | Design can be completely removed | SAFETY | Maneuverable |
| 3 | The sample collected can be varied | | 30mg - 80mg sample collection |
| 4 | The design is able to report current stage in it's process | | Display feedback |
| 5 | Should be modular/lightweight | | Weight (lbs) size ($cm^3$) |
| 6 | Necessary power sources should be included | | Batteries/Compressed Air/Rat |
| 7 | Is a novel creative solution | | Unique (by comparison) |
| 8 | Device is reasonably fast | | Time (s) |

## 2.2 Technical and User Benchmarking

The team performed technical benchmarking by researching existing products that satisfy the interpreted needs, as shown in Table 2.2-1. The group chose to use devices that remove material from narrow passages instead of solely focusing on devices used to collect samples from pressure tubes in reactors. This was due to the innately confidential nature of tooling such as CWEST and the need to try and maintain a creative outlook for the project, as the client strongly suggested creativity as a desirable objective.

From the product benchmarking, a notable feature of the CWEST device is that it is controlled and powered by the BRIMs system, which, for the project, we will not have access to. Including

the Mars Rover in the product benchmarking provides many tangible features to our potential design objective. Other devices were chosen because they meet at least one design criterion.

Table 2.2-1: Product Benchmarking

| Device Specifications | Device | CWEST | Arterial Catheter | Drain Snake | CSA420 Endress | Mars Rover - Curiosity |
|---|---|---|---|---|---|---|
| Company | | Kinectrics | Generic | Generic | Endress & Hauser | NASA |
| Cost | | $$$$$$$$$$ | $ | $ | $$$ | $$$$$$$$$$$$$ |
| Stability | | BRIMs | Guided by artery | None | | Multiple surface contact points |
| Shape | | Tube | Wire | Wire | Tube | Rectangular |
| Power | | External: BRIMs | N/A | External: Man power | External: Man power | Solar + Radioisotope |
| Failsafe | | BRIMs | Pressure Drop | None | None | Safe-mode software, sensors |
| Controls | | BRIMs | Direct hand | Direct Hand | Control panel | Digital signalling |
| Sampling Method | | Scraping | Suction | Friction | Water Pressure | Drill, Scoop |
| Size | | ~4 ft | 20 gauge | 35.5” x 0.45” | 381 mm x 75 mm | 10’ x 9’7” x 7’ |
| Sample Size | | 30-80mg | 1 mL | N/A | Liquid | 45mm³ |
| Material | | Zirconium | Polyurethane, Silicone, Nitinol, Stainless Steel | plastic, metal | Polyamid PA, Stainless Steel | 7075 Aluminum, Titanium. Aerogel |

Task-based user benchmarking was conducted using raw data from the client meeting. This benchmarking helps the team see where each need has its practical situation. The client mentioned overwhelming positivity towards the CWEST device and had no notable changes to note on an empathy map. Additionally, there is insufficient information to create a comprehensive journey map. User benchmarking can also be conducted later in the project to develop a feedback system for our prototypes. This will eventually give a more holistic framework by which we can gauge how users may react to the developed design and act as an effective communication tool for the presentation.

Table 2.2-2: User Benchmarking - Tasks of CWEST

| Task | Concern |
|---|---|
| Enter Reactor vault | Can the device fit in the vault |
| Attach CWEST device | Weight and maneuverability |
| Operate CWEST device | Sufficient power, damage to pressure tube |
| Collect Sample | Exposure to sample |
| Detach CWEST device | Exposure to contaminated tooling |
| Analyze Sample | Sample quantity |

## 2.3 Engineering Design Specifications

Design specifications were developed and organized into three categories based on the design criteria highlighted in Table 2.1-1. These categories include functional requirements, constraints, and non-functional requirements. These requirements will be used to evaluate the ideations and prototypes that will be developed.

Table 2.3-1: EDS - Functional Requirements

| Design Specifications | | Relation (=, < or >) | Value | Units | Verification Method |
|---|---|---|---|---|---|
| **Functional Requirements** | | | | | |
| **1** | Collect a metal sample | = | 30-80 | mg | Test |
| **2** | Completely Removable | = | Yes | N/A | Test |
| **3** | Enclosed Sample | = | Yes | N/A | Test |
| **4** | Stage Reporting | < | 2 | s | Test |
| **5** | Visual for process | = | Yes | in, dpi | Test |
| **6** | Device speed (time) | < | 10 | min | Test |
| **7** | Power source included | = | Yes | N/A | Analysis |
| **8** | Distance Traveled | = | 15 | ft | Test |

Table 2.3-2: EDS - Constraints

| Design Specifications | | Relation (=, < or >) | Value | Units | Verification Method |
|---|---|---|---|---|---|
| **Constraints** | | | | | |
| **1** | Weight | < | | kg | Analysis |
| **2** | Size (Diameter) | < | 4 | in | Analysis |
| **3** | Size (Length) | < | 4 | ft | Analysis |
| **4** | Sample Exposure | < | Yes | N/A | Test |
| **5** | Sample Collection | = | 30 - 80 | mg | Test |

Table 2.3-3: EDS - Non-Functional Requirements

| Design Specifications | | Relation (=, < or >) | Value | Units | Verification Method |
|---|---|---|---|---|---|
| **Non-Functional Requirements** | | | | | |
| **1** | Aesthetics | = | Yes | N/A | Survey |
| **2** | Safety | = | Yes | N/A | Analysis |
| **3** | Corrosion Resistant | = | Yes | N/A | Analysis |
| **4** | Operation Time | < | 10 | mins | Test |
| **5** | Creative Solution | = | Yes | N/A | Analysis |

# 3.0 Conclusion

## 3.1 General Progress

Sample Snatchers has established a structured list of design criteria, prioritizing safety, removability, and maneuverability. The engineering design specifications tables include at least one design criterion for each identified need. Benchmarking had no immediate significant impact on the previously completed Deliverable B but does give a frame of reference for future concepts. From benchmarking, the components required for a functioning device can be determined. It has also allowed different sampling methods and their implementation to be considered. The group has also made progress in developing ideations for subsystems for use in Deliverable D. Future work will be able to include a validation process for developed designs by using the content of this report.

## 3.2 Client Meeting Reflection

As the team developed the design criteria, the initial client meeting provided useful communicational indications to guide the ranking of these criteria' relative importance and create a problem statement. In Deliverable B, the following sequential process was performed: collecting and interpreting raw data from the client meeting, formulating critical user statements, transforming user statements into interpreted user need statements, and establishing importance. Based on the hierarchy of the interpreted needs, the developed design criteria provide descriptive guidance of what the product must be.

## 3.3 Trello: Task Plan Update

In Trello, the team's task board has been updated to include changes in task duration, missing tasks, task responsibilities, and further details to maintain effective project management. The group has begun on Ideations to develop concepts for Deliverable D.

Please refer to the following link to review updates:

https://trello.com/invite/b/67846e4e793783740d87aed9/ATTIbb8224a0c532547c0a69acd41093c7c7A4D60EC7/gng1103-project-sample-snatchers

# Citations


Amazon.com. (2025, February 1). *Customer reviews for 35.5 inch Drain Clog Remover.* Retrieved February 1, 2025 from https://www.amazon.com/product-reviews/B0C3L7FKCQ/ref=cm_cr_arp_d_viewopt_kywd?filterByStar=five_star&reviewerType=all_reviews&pageNumber=1

Cotterill, P. (n.d.). *The hydrogen embrittlement of metals*. Progress in Materials Science, 9(4), 205-250. https://doi.org/10.1016/0079-6425(61)90005-6

KINECTRICS (2021, January 18) Fuel Channel Inspection Tools. Youtube https://www.youtube.com/watch?v=J-Pnytp9b2g&t=0s

Custom circumferential wet scrape tool (CWEST) Design & Supply. Kinectrics. (n.d.). https://www.kinectrics.com/projects/updated-design-supply-of-a-custom-circumferential-wet-scrape-tool

Fuel Channel Inspection & Maintenance. Kinectrics. (n.d.-b). https://www.kinectrics.com/capabilities/products/nuclear-generation-products/fuel-channel-inspection-maintenance

Assessment and management of ageing of major nuclear power plant components important to safety: Candu Reactor Assemblies. (n.d.). https://www-pub.iaea.org/MTCD/Publications/PDF/te_1197_prn.pdf